As a final example, let's consider the task of enumerating all the couples of numbers in a list, regardless of their order. For instance, given the list `{1, 2, 3}`, we have the pairs `(1,2)`, `(1,3)`, `(3,1)`.

Turns out, the number of unordered pairs for n elements is $T_3(n) = n * (n-1) / 2$.

$$T_3(n) = n * (n\text{-}1) / 2 \leq 1 * n^2 \quad \text{for} \quad n > 1$$

and thus we can say $T_3(n) = O(n^2)$. This time, the quadratic bound is also a strict bound as shown in figure B.4.

[1] This might be useful if we need to prove that an algorithm will actually run in finite time. But with real-world applications, it's usually of little practical use.